

Duy Nguyen

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Education

California Institute of Technology (Caltech)

Sep 2022 – Jun 2026

B.S. in Computer Science & B.S. in Information and Data Science

GPA: 4.1/4.0

Relevant Coursework: Statistical Foundations of Machine Learning, Quantitative Risk and Portfolio Management, Machine Learning & Data Mining, Statistical Inference, LLMs for Reasoning, Computation Theory, Algorithms

Technical Skills

Languages/Frameworks: Python, PyTorch, TensorFlow, LangChain, LangGraph, C/C++, R, Java, OCaml, Haskell

Statistical & ML Methods: Generative modeling (diffusion, flow matching), time-series modeling, self-supervised learning, statistical inference, backtesting, Monte Carlo simulation, SHAP

Tools: Linux, FAISS, Snowflake, Git

Professional Experience

Google — Software Engineer

Sep 2026 – Present

Trusted Partner Cloud Infrastructure

- Building change-management and compatibility infrastructure for deploying Google services across isolated cloud environments

NVIDIA — AI Applied Research Intern

USD Smart-Materials

Jun 2025 – Sep 2025

- Built a generative AI pipeline (**SDXL, FLUX diffusion/flow-matching models + ControlNet**) that produced photorealistic renderings, significantly boosting a Vision-Language Model's material-prediction accuracy on CAD inputs
- Proposed and prototyped a video-based approach using NVIDIA's **COSMOS** model to produce temporally coherent, multi-view-consistent outputs for downstream VLM analysis

USD Code

Jun 2024 – Dec 2024

- Built a synthetic data-generation pipeline (**LangChain, GPT-4, LLaMA3**) using a persona-driven methodology, expanding NVIDIA's proprietary LLM training set **10x** and lifting eval accuracy **10%**
- Improved answer quality by building a RAG pipeline, fine-tuning LLaMA3, and integrating a FAISS vector database

Stratus Data — AI Applied Research Intern

Mar 2025 – Jun 2026

- Architected a multi-agent data-analysis system (**LangGraph/MCP**) where a supervisor LLM delegates to specialized agents, including a Cortex Analyst agent querying a Snowflake database directly from natural language, to automate business-insight discovery

Google — STEP Intern

Jun 2023 – Sep 2023

- Fixed bugs and extended a server-side Google Play debugging feature, giving developers visibility into app, item/non-item card, and Play Store stream diagnostics

Research Experience

Caltech — Undergraduate Researcher

Jun 2024 – Jun 2026

Anima AI + Science Lab

- Developed a resolution-invariant time-series framework (Flow Matching + STFT/Superlet Transform) supporting multi-level supervised learning, outperforming masked autoencoding and Fourier Neural Operator baselines on neural decoding, sleep-stage classification, and activity recognition (**accepted: NeurIPS 2025 workshop, TMLR**)
- Improved performance **20%** via grid search over feature-extraction windows and layer configurations

Yisong Yue Lab & Eva Dyer Lab — Caltech + Georgia Tech

- Designed a novel positional encoding for self-supervised, masked-autoencoding transformers on 1D neural time-series data, removing reliance on 3D electrode locations; boosted downstream classification/decoding **15%** over supervised baselines (**accepted: NeurIPS 2025 workshop, Spotlight**)

Projects

FGNO Signal Research for Equity Alpha

- Pretrained an FGNO Transformer backbone via self-supervised flow matching on STFT spectrograms of daily equity price/volume data; probed frozen representations across 8 downstream regressors and a 36-way (layer, flow-time) grid to predict 5-day forward returns
- Identified an apparently profitable mean-variance-optimized long/short signal (pooled Sharpe +2.07), then rejected it after diagnosing failure under Newey-West/bootstrap significance correction for overlapping-label autocorrelation and significant losses (Sharpe -7 to -10) under walk-forward re-validation

Trajectory Flow Matching for Financial Time-Series Modeling

- Adapted Trajectory Flow Matching, a stochastic bridge model with a causal Transformer history encoder, to estimate the joint next-day distribution of equity returns, volatility, abnormal volume, and liquidity via Monte Carlo scenario generation
- Evaluated Neural SDE, probabilistic Transformer, and LightGBM quantile baselines across four expanding-window folds using energy score, QLIKE, and rank IC; implemented dollar- and sector-neutral long/short backtests with turnover, transaction costs, market impact, and borrow fees